WARG ROBOTICS



REAL WORLD, UNREAL EXPERIENCES

A camera you can build on.

Low-latency digital video, onboard recording and MAVLink telemetry on one encrypted link, with a companion computer on the way.

GLASS TO GLASS	RANGE	BAND	ENCRYPTION
60 ms	5 km	5.8 GHz	256-bit

IN THE FIELD

Three jobs it does today

What we have proven on each, and what is still ahead of us.



FPV and freestyle

Below about a hundred milliseconds the link stops being something a pilot compensates for.

- 60 ms through gates and tight lines
- Holds under hard banking and prop wash



Computer vision

For anything reactive, like obstacle avoidance or target tracking, round-trip latency decides whether it works.

- Stable link at extended standoff
- Onboard compute with the Warg SDK



Remote cockpit

Range and encryption matter most when the asset is far from the operator and the footage may be evidence.

- Survives sustained chassis vibration
- Low mounting height

BUILT AROUND YOUR CONSTRAINT

Imaging to your optical path

Lens and field of view chosen per airframe; the sensor swapped where the light demands it.

Chassis cut to your vehicle

Mount pattern, stack height, connector orientation and weight budget agreed before tooling.

Transport tuned to your site

Channel plan, bitrate and range matched to your link budget and regulatory ceiling.

An open feed

Standard output into your own pipeline. No proprietary decoder in the path.

CHOICE ONE

The camera

Three units on one link. They differ only in the picture and the storage.



Ghost JON

AVAILABLE NOW

Live and recording	720p · 60 fps
Sensor	8 MP Sony IMX219
Storage	8 GB ~50 MIN AT FULL BITRATE
Latency · range	60 ms · up to 5 km



Greywind ROBB

AVAILABLE NOW

Live and recording	1080p · 30 fps 720P AT 60 FPS
Sensor	8 MP Sony IMX219
Storage	16 GB ~1 H 45 AT FULL BITRATE
Latency · range	60 ms · up to 5 km



Raven BRAN

LAUNCHING SOON

Recording	4K · 30 fps PROVISIONAL
Storage	32 GB PROVISIONAL

SHARED BY EVERY UNIT

The link

A two-part 5.8 GHz digital link carrying video and two-way telemetry at the same time. Quality adapts to the signal, so the picture degrades gracefully at range instead of cutting out.



Frequency band	5725–5875 MHz
Channels	149, 153, 157, 161, 165, 169, 173
Channel width	20 MHz 40 MHZ ON 149, 157, 165, EXPERIMENTAL
Modulation	OFDM (802.11n-based), BPSK to 64-QAM
Antennas	2 per unit, 2T2R MIMO with diversity
Transmit power	1–25 dBm, adjustable DEFAULT 5 DBM
Video bitrate	1.5–19 Mbit/s, adaptive
On link loss	Drops to its most robust mode, recovers without a restart
Security	256-bit authenticated encryption VIDEO, TELEMETRY AND CONTROL
Pairing	One key per air and ground pair

The region setting enforces local channel rules; India is the default. The operator sets power to suit the antenna and stays within local EIRP limits (India: 36 dBm, 4 W).

INTEGRATION

Into your stack

The air unit takes the camera and your flight controller; the ground unit drives the screen and serves your ground station. Both run from 5 V.

	AIR UNIT	GROUND UNIT
Role	Camera, encoding, recording, transmitter	Receiver, display, setup hotspot
Camera input	MIPI CSI-2, camera included	—
Flight controller	UART, 3.3 V logic, to 921600 baud	—
Video out	—	HDMI, with OSD
Ground control	—	MAVLink over UDP, Wi-Fi or Ethernet
Local network	2.4 GHz Wi-Fi, off for flight if you want	2.4 GHz hotspot (WPA2), Ethernet
Antennas	2 × 5.8 GHz	2 × 5.8 GHz
Supply	5 V; radio on its own feed recommended	5 V
Thermal	Automatic protection, three stages	Automatic protection

WORKS WITH

Flight controllers	ArduPilot, PX4, INAV (MAVLink)
Ground control apps	QGroundControl, Mission Planner, any MAVLink app
Ground control access	Monitor-only by default; full control in settings
Display	Any HDMI monitor or HDMI goggles
Setup	Browser page on the ground unit; no app
Updates	Over the air, signed and checked before install

CHOICE TWO, INDEPENDENT OF THE FIRST · LAUNCHING SOON

The compute

Every unit in the line will be available as a Pro: the same camera and the same link, with a Linux processor on board. Memory and storage are chosen separately from the camera.

MEMORY	STORAGE
2 / 4 / 8 GB	32 / 64 / 128 GB

WHAT YOU CAN RUN ON IT

Image processing

Detection and tracking against the full-rate frame, where it is captured.

Autonomous navigation

The aircraft acts on what it sees, so the link is no longer in the control loop.

Path finding

Routing computed from the live picture rather than a map made earlier.

Precision flight

Close-quarters work where ground-station latency is the limiting factor.

PLATFORM

Processor · OS	ARM 64 · Linux
Software	Warg SDK, container deployment
Protocol · connector	MAVLink · JST-SH
Input voltage	5.5–36 V

DETAIL

Video and recording

720p at 60 fps in H.264 by default. Recordings are written on the vehicle at the live-stream quality.



VIDEO

Default mode	1280 × 720, 60 fps
Selectable	1920 × 1080, 1640 × 1232, 1280 × 720, 1024 × 576, 960 × 720, 854 × 480, 640 × 480
Codec · bitrate	H.264 · 1.5–19 Mbit/s, adaptive
Sensor · field of view	8 MP Sony IMX219 · 140 deg diagonal
Image controls	Rotation 0/180°, noise reduction, sharpness, contrast, saturation, brightness, ±4 EV, white balance presets or manual

RECORDING

Format	MP4 (H.264), as the live stream
Control	From the ground unit; state on the OSD
File length	A new file every 10 minutes
Storage protection	Stops cleanly below 300 MB free
Download	Over Wi-Fi, from the ground unit's page

Get in touch

Tell us what you are building and we will take it from there. Bulk orders, OEM programmes, custom work and evaluation units all start the same way.



BUYING TODAY

Ghost	₹29,999 + GST
Greywind	₹35,999 + GST
Ghost + Greywind	₹49,999 + GST per pair
Pro, Raven	Launching soon; quoted on the configuration

Buying for a business? Add your GSTIN at checkout and the GST invoice carries it, for input tax credit. Up to 9 units per online order; for more, write to us.

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WARG ROBOTICS

DESIGNED AND MADE IN INDIA